Minors / guess-and-determine ECDLP vs matched negation rho: panel relaunch C

Study `research/minors_guess_determine_20261107` · run `minors-gd-panel-20261108-c`, pod uid `b8617fcd-e2e5-4235-90b0-b201e5bb3857` · 2026-11-08 · every number is **[pcc:unverified]**. The numbers come from `panel/analysis.json`, the frozen `minors_gd_analyze.py` output and the primary analysis. An independent re-derivation agreed on all 1017 compared quantities, with bootstrap bounds equal to Monte-Carlo precision. Full report: `RESULTS.md`.

Verdict (PROTOCOL.md section 8, mechanical)

CLOSED for the reconstruction: the six arms `mgd.all2`, `mgd.gesc`, `mgd.apm2`, `gd`, `gd_dp1` and `gd_half`, in prime fields at 20-40 bits.

- Every arm has upper95(e_all) ≥ 0.5. The values run from 0.815 to 1.065, and every lower bound is ≥ 0.766.
- Every arm costs more than rho at every size: the minimum ratio is 4.37.
- No real-lead flag is raised.
- The ordering holds at inversion weights w = 1.5 bits(p), 3 and 10, and at unpriced weights λ = 0, 0.1 and 1, so the result is not unordered.

This is not an ecbench session, and it is not a leaderboard row.

Distance remaining to matched rho (cheapest minors arm per size, paired bootstrap 95%)

bits	20	24	28	32	36	40
rho cost (mults)	1.09e5	1.91e5	3.67e5	9.95e5	3.48e6	1.29e7
cheapest arm	`mgd.gesc`	`mgd.gesc`	`mgd.apm2`	`mgd.apm2`	`mgd.apm2`	`mgd.apm2`
its cost (mults)	4.78e5	1.17e6	7.10e6	8.15e7	1.27e9	2.00e10
ratio to rho	4.37	6.11	19.3	81.9	365	1552
95%	4.28-4.46	5.75-6.51	17.2-21.7	71.8-93.1	309-431	1049-2721

Exponents (slope of log2 cost per bit; bootstrap 95%)

arm	**e_all (primary)**	**e_top3 (32-40)**	**e_arm - e_rho, all sizes**
`rho`	0.346 [0.342, 0.349]	0.461 [0.447, 0.476]	-
`mgd.all2`	0.890 [0.853, 0.968]	0.969 [0.842, 1.249]	[0.507, 0.622]
`mgd.gesc`	0.823 [0.798, 0.861]	1.031 [0.942, 1.170]	[0.451, 0.517]
`mgd.apm2`	0.787 [0.766, 0.816]	0.992 [0.919, 1.093]	[0.420, 0.471]
`gd`	0.835 [0.786, 0.922]	n/a (<3 sizes)	[0.440, 0.577]
`gd_dp1`	0.979 [0.919, 1.065]	n/a (<3 sizes)	[0.574, 0.720]
`gd_half`	0.930 [0.838, 1.005]	n/a (<3 sizes)	[0.493, 0.659]

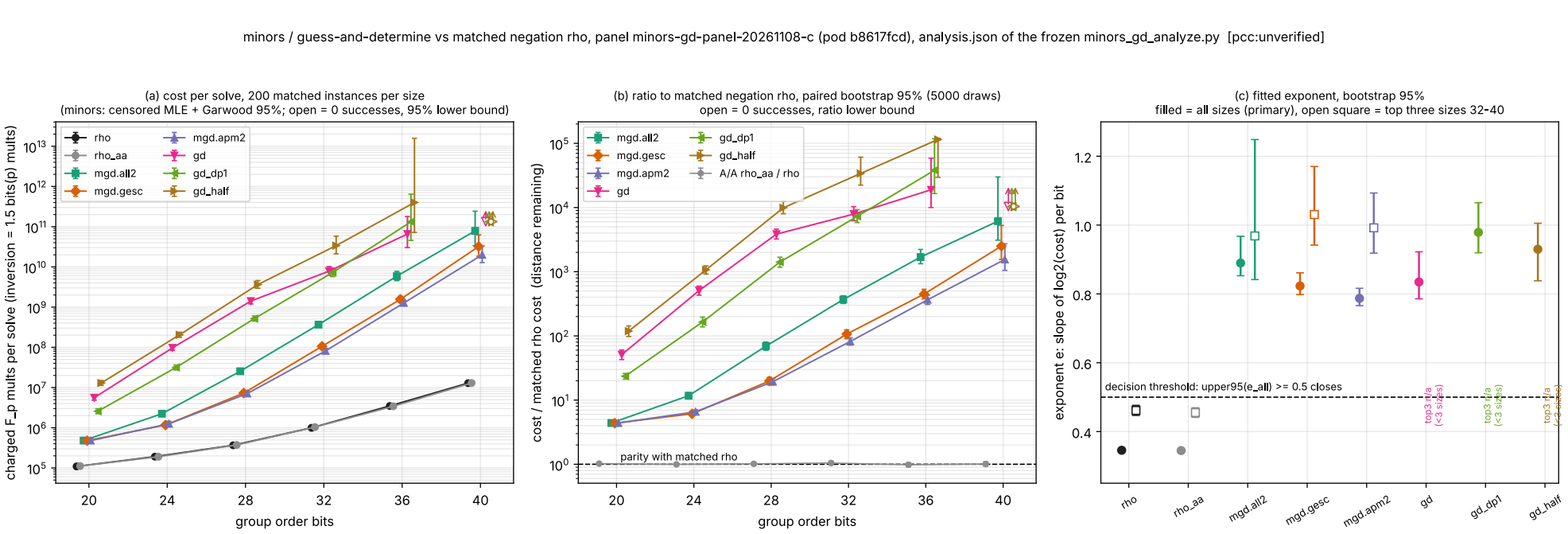


Figure 1: Values from `analysis.json` of run `minors-gd-panel-20261108-c`, with 200 matched instances per size. (a) Cost per solve, 95% intervals; open markers are zero-success lower bounds. (b) Ratio to matched rho, with the A/A ratio. (c) Exponents against the 0.5 decision threshold.

Controls

- **Positive.**
  - Development self-test: all arms listed recover k ("SELFTEST PASS"). `gd_dp1` and `gd_half` are not in the self-test list.
  - In the panel, all 6956 successes are correct.
- **Negative (a).** The verifier rejects k+1 (development).
- **Negative (b).** Curve vs uniform zero-minor counts agree at 12-16 bits (development only).
- **Negative (c).** `bad_candidates` = 0 in all 9600 records.
- **A/A.** rho_aa/rho has every 95% interval containing 1, with half-width 2.6-8.0%.

Execution record

- **Launch 1** (14 vCPU) was never scheduled: 14180m exceeds the 14000m allocatable. It was deleted with no data.
- **Relaunch B** (12 vCPU) was lost to Spot (exit 137, node shutdown). It is not admitted.
- **Relaunch C**, the single allowed relaunch, ran 12:46:21-16:45:14Z, exit 0 `Completed`.
  - The exit comes from the EKS audit log (audit id `c22e0d20-...`), because the Pod object was deleted with its node.
  - JOBS 14 to 12 changes only the shard parallelism: the measured quantities are exact counts.
  - `SHA256SUMS` verifies.
  - The protocol, run_panel and analyze hashes equal the frozen values.
- Wall time (about 47.8 vCPU-h) is informational. Total charged work is 6.69e12 units against the 7e12 the protocol estimated.

Caveats

- This is a reconstruction and not the authors' code. It runs in prime fields, without `apm3`.
- The rho in PROTOCOL section 2 text (R = 64, 32 walkers) is stale. The frozen `rho.rs` has R = 32, W = 16.
- The censored MLE assumes a constant hazard. `gd*` runs a single attempt per run from 28/32 bits up.
- Sparse successes affect the fits: `gd_half` has 1 success and `gd_dp1` 3 at 36 bits, which makes some bootstrap draws drop a size.
- There is no top3 exponent for the `gd` family.
- The protocol's "per-attempt estimate in the README" check is UNRESOLVED as written: the README has none.
- The README's "five orders for gd at 40 bits" is UNRESOLVED: only a lower bound (> 1.04e4) is available.
- There is no replay certificate, validator report or ecbench cross-check.